

WATERWALKER · DESIGN DECISIONS REGISTER

WaterWalker design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WWK-DEC-001	Version	0.1
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwalker		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-02	Amish Chadha	Register opened with the design made constructable (WWK-DDR-003); budget treated as a value-engineering target

WaterWalker design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under docs/decisions/ ; this register is the index Amish works from. The build plan (docs/05-build-plan.md , WWK-BLD-001) describes the design as it stands and does not list open decisions.

■ **Open decisions**

Table 1. Open decisions, Proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P16	Accept; accept with changes; return to the concept layout	Accept: every change keeps what the carrier does and is needed for it to be built	The whole build plan	WWK-DDR-003, Table 1
2	R2 (firm path, 60.8 N against 60 N) and R4 (10 % climb, 181.5 N against 180 N) now not met, because the constructable carrier is 2.2 kg heavier	(a) accept on paper and confirm the push limits with users in co-design; (b) apply the open mass options: tires without a puncture belt, no liners (1.6 kg; R2 60.0 N, R4 179.2 N); (c) 1.2 mm wall for the bearers and rear brackets (about 0.5 kg)	(a), keeping the puncture protection chosen on 2026-09-25	Tires and liners (BOM line 14) or the carrier tube	WWK-DDR-003, A1
3	Ground clearance under the cradle carrier: 125 mm (the concept floor was 150 mm up, with nothing under it)	(a) accept; (b) raise the cradle 25 mm (lift height 389 mm, centre of mass 25 mm higher)	(a); set a clearance requirement after the first route survey	Hanger lengths, cradle height	WWK-DDR-003, A2
4	First co-design partner and region	Helpful Engineering network, an NGO or a university in a region with sandy routes	None; partners are picked per area later	None in this build; sets the push-force limits, routes and pot sizes the build is tested against	WWK-DDR-001 item 7
5	Dead-man brake	Ask users in co-design; a spring-applied brake released by the hip bar; or rely on the lever, latch and lock pin	None until users are asked	Brake lever and cables	WWK-DDR-001 item 8
6	Value-engineering target for the later assist prototype	Kit about USD 260 with one motor, about USD 450 with two, plus a shared SwapCell pack	None	Not part of the first build; the mounting points are	WWK-DDR-002 item 14
7	Weld fatigue at the frame joints	Gussets at the rear cross member and caster arm joints; a local 1.5 mm wall; accept until a fatigue test	Accept until a TRL 4 fatigue test: the stress range in the rail is now 61 MPa, below the 71 MPa reference, with the cradle hung from the cross members	Frame weldment	docs/REVIEW.md (2026-09-25); WWK-CAL-001 v0.3 section 7
8	Appearance-model choices from the product renders: grips at the front of the grip tubes, inboard grips on the	Adopt each or not	Adopt the cable routing, hand holds and plates; ask users about grip position and spacing at	Grip tubes, cradle walls, labels	docs/REVIEW.md

#	Decision needed	Options	Recommendation	Affects in the build	Source
	hip bar, left brake cable along the hip bar, hand holds in the cradle side walls, the rated load and slope plate on the right rail and a second inside the left rail		the first co-design session		(2026-09-26), items 1 to 5

■ To confirm when parts are bought

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	The rigid 26 in forks: steel, 1 1/8 in steerer at least 170 mm above the crown, 100 mm dropouts, and their offset (the model uses 60 mm)	The offset is the caster trail and sets where the rear head tubes sit; steel blades take the welded tabs	WWK-DDR-003, P3, P13
2	External-cup (EC34) headsets fit the 34 mm head tube bore	The head tubes are cut and reamed to suit	WWK-DDR-003, P7
3	100 mm front-type drum hubs are sold and serviced in the region, and their reaction arm and clip fit the fork blade	R12; the reaction arm must lie along the blade	WWK-REQ-001 R12; WWK-DDR-003, P9
4	The brake lever with parking latch clamps on 22.2 mm tube and a cable splitter feeds both drums	The grip tubes are sized for it	WWK-DDR-003, P8
5	A 10 mm ball-lock pin with about 140 mm usable length	It must reach the outer tab and stay inside the axle nuts	WWK-DDR-003, P10
6	The tires chosen (2.1 to 2.4 in) clear the fork crown by 25 mm or more	The model has 33 mm with 2.1 in tires	<code>cad/src/model.py</code>
7	The real diameter of the clay pots users carry	Two 380 mm pots fit with 12 mm to spare (R1)	WWK-REQ-001 R1

■ Value engineering

Value-engineering target: USD 450 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 462 for the first prototype without assist and with mounting points (USD 12 over the target). Main cost drivers and savings worth trying:

- The largest lines are the wheels and forks (USD 168 for four built wheels in forks, BOM lines 2 and 3), the tires, tubes and liners (USD 96, line 14), the frame weldment (USD 61, line 1) and the four headset and steering lock sets (USD 32, line 16).
- Making the design constructable added USD 36: the four headset and lock sets (line 16, USD 32, of which USD 12 was in line 3 before), the cradle carrier and second rear brackets (line 1, USD 6), cradle brackets and bolts (line 5, USD 5), guard clips (line 8, USD 2), the second lock pin tab (line 15, USD 1) and fixings (line 12, USD 2).

- Savings worth trying: plain bushed rear head tubes held by a cross bolt instead of rear headsets (about USD 10, but two kinds of head tube to make); buying the four wheels as one built set (often USD 10 to 20 less than four singles); tire liners only on the rear wheels (USD 8), if co-design shows punctures are rare on the routes.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 6: 26 in wheels; no assist in the first prototype, with mounting points; full-swivel lockable casters; user between the rear wheels; four jerrycans or two clay pots; rear drum brakes with a parking latch	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	WWK-DDR-001
2026-09-25	Cross-cutting: assist mounting points to SwapCell interface v0.3 (items W and V); SwapCell pack priced once in the SwapCell BOM; co-design partners picked per area later	Amish, same instruction	WWK-DDR-001
2026-09-25	Items 9 to 13: rear wheels in fixed forks with 100 mm drum hubs and a 770 mm track; risers at 1,150 mm and a 1,530 mm wheelbase; 40 x 30 mm main tube; 1.2 mm main wall and thinner cradle, puncture protection kept, R9 revisited with users; positive parking lock pin	Amish: "i accept all your recommendations, go with them across all repos."	WWK-DDR-002
2026-09-30	Build plan format with pictures for every component and step; open decisions kept out of the build plan and in this register; the design made physically buildable as the pictures are drawn	Amish: "fix the design assumptions to match and be physically feasible as you draw the illustrations" and "don't log outstanding decisions in this build plan"	WWK-DDR-003 (changes open for review, Table 1 item 1)
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	<code>.kit/STANDARDS.md</code> section 18